Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

4DDAPM Module: -

The RS485 IO module is a compact, robust, and low-power device designed for reliable communication over distances up to 700 meters without a repeater, depending on the signal strength of the Modbus master. It features 4 digital inputs, 4 digital outputs, 4 analog inputs (10-bit resolution, supporting 0–20 mA and 0–10 V), and 4 RTD (PT1000) channels, making it ideal for versatile industrial applications. Housed in a DIN rail-mountable enclosure, the module includes exposed connectors for easy wiring and LED indicators for real-time I/O status monitoring. Configuration of slave ID and baud rate is done via internal DIP switches. To enhance system protection, the module is equipped with resettable fuses guarding against reverse polarity on both power and RS485 communication ports.

Specifications

General –

I/O Connectors	30 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	RoHS, CE (APPROVED)

DI Inputs –

Channels	4
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic ‘0’ = <1 VDC, Logic ‘1’ = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Digital Output

Channels	4
Output Type	P-Channel MOSFET
Channel Voltage	12 VDC
Channel Current	20A
Channel Power	240 W (max)
Operating Time	2.8ns.
Turn Off delay Time	33.5ns.
Operating Temperature	-55℃~+150

AI Inputs –

Channels	4
Input Signal	4 – 20 mA / 0 – 10VDC
Accuracy	± 2% Full scale
Input Resolution	10 bits
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

AI Inputs –

Channels	4 – PT1000 – (jumper selectable) Three wire
Accuracy	Total Accuracy over All Operating Conditions: 0.5NC (0.05% of Full Scale) max.
Input Resolution	15-Bit ADC Resolution.
Isolation	Optically Isolated.

Additional Features: -

All inputs and communication ports isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

Configuration Settings: -

Communication Speed	9600 – 115200 (DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable with DIP Switch
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3.
Function code DI	0x02 Read discrete Input
DI Register Address	4,5,6,7.
Function Code AI	0x03 Read Holding Registers
AI Register Address	10 Bit - 1,2,3,4.
RTD Register Address	5,6,7,8.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	**DO 1**	00001	0X05,0X0F	RS485	1 Bit Boolean
2	**DO 2**	00002	0X05,0X0F	RS485	1 Bit Boolean
3	**DO 3**	00003	0X05,0X0F	RS485	1 Bit Boolean
4	**DO 4**	00004	0X05,0X0F	RS485	1 Bit Boolean
5	**DI 1(With & Without Potential)**	10005	0X02	RS485	1 Bit Boolean
6	**DI 2(With & Without Potential)**	10006	0X02	RS485	1 Bit Boolean
7	**DI 3(With & Without Potential)**	10007	0X02	RS485	1 Bit Boolean
8	**DI 4(With & Without Potential)**	10008	0X02	RS485	1 Bit Boolean
9	**AI 1**	40002	0X03	RS485	16 Bit Unsigned int
10	**AI 2**	40003	0X03	RS485	16 Bit Unsigned int
11	**AI 3**	40004	0X03	RS485	16 Bit Unsigned int
12	**AI 4**	40005	0X03	RS485	16 Bit Unsigned int
13	**RTD1 1000**	40006	0X03	RS485	16 Bit Unsigned int
14	**RTD2 1000**	40007	0X03	RS485	16 Bit Unsigned int
15	**RTD3 1000**	40008	0X03	RS485	16 Bit Unsigned int
16	**RTD4 1000**	40009	0X03	RS485	16 Bit Unsigned int

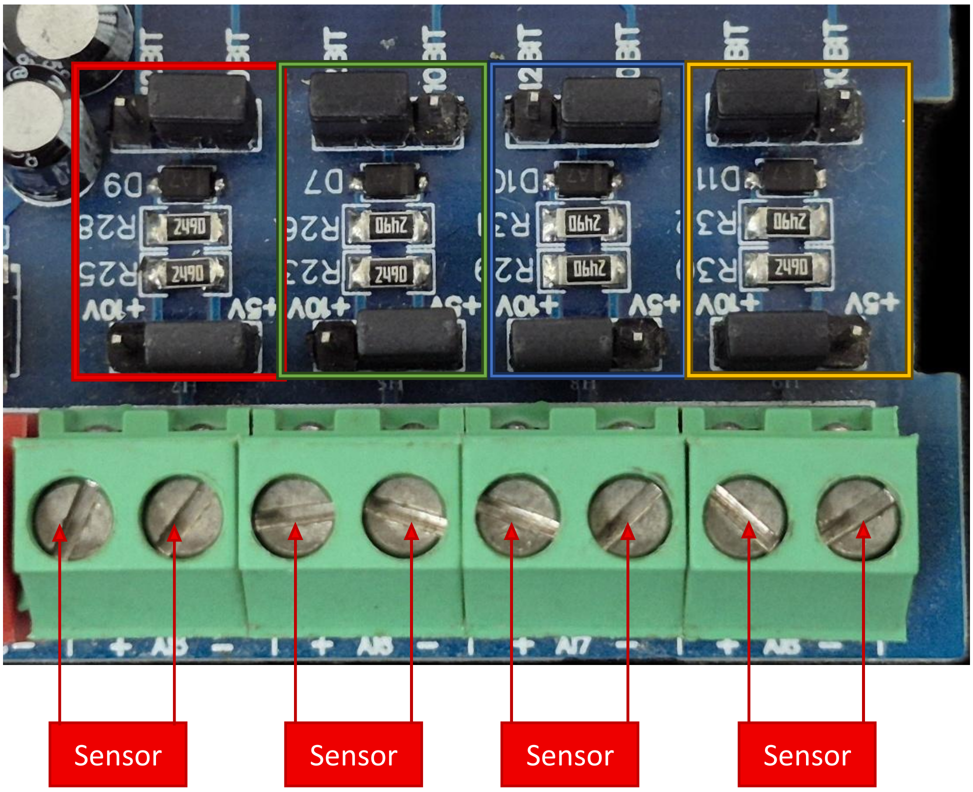
Note: -

For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

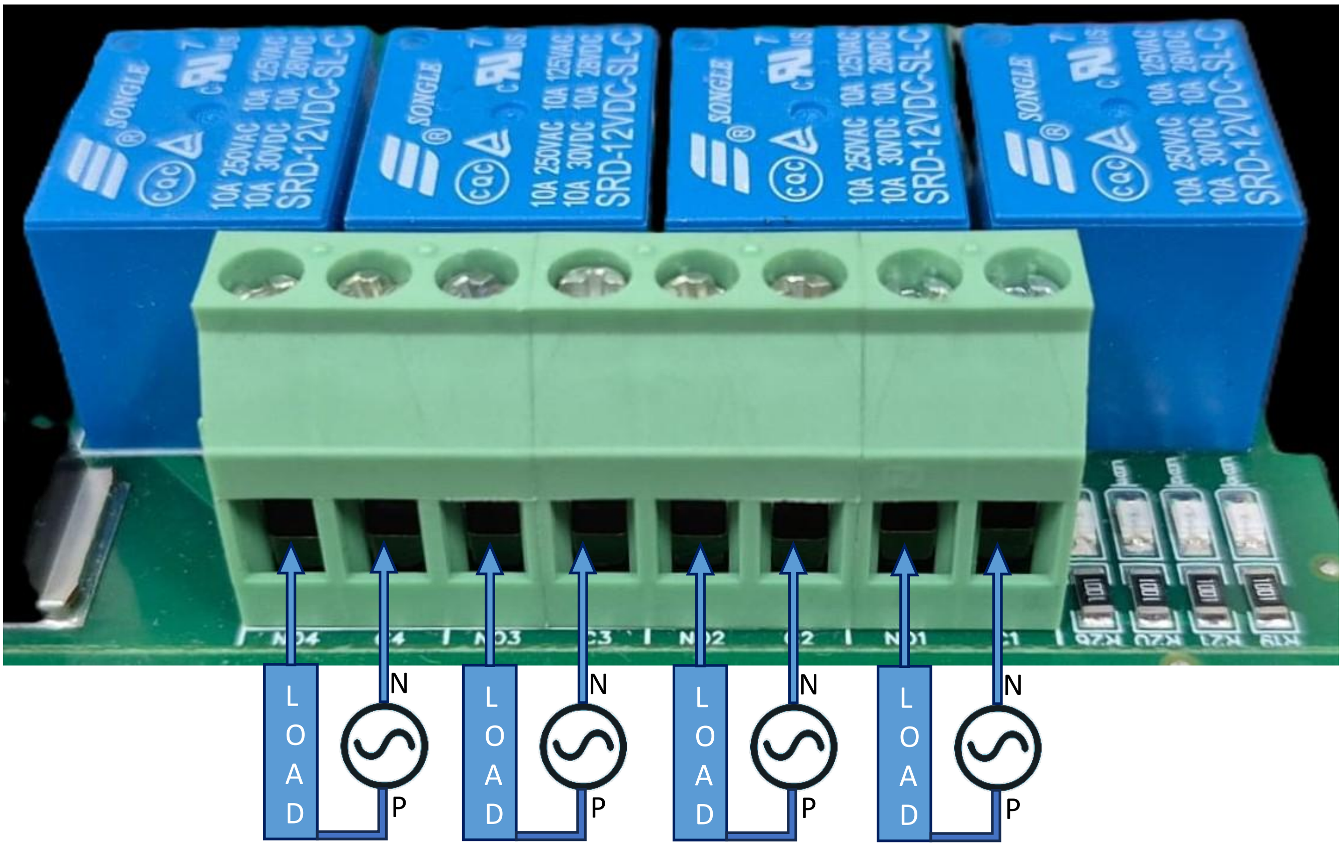
22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: **+** (Positive) and **-** (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
  - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
  - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
  - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
  - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
  - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
  - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.

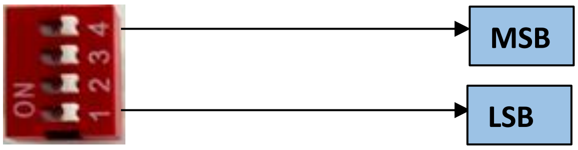
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C is internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO remains open**, and the load stays OFF.
- **Load Wiring Connections:**
    - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
    - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

  <u>Note:</u> Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or 10A @ 125VAC) or **10A @ 30VDC** (or 10A @ 28VDC).

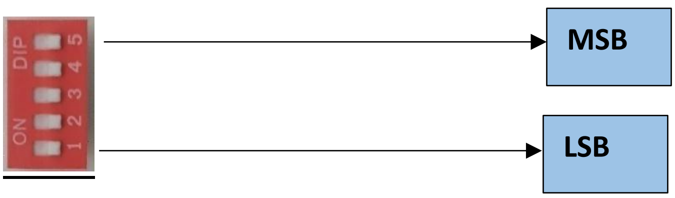
BAUD RATE DESCRIPTION



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition, pls press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
115200	ON	OFF	OFF	OFF
57600	OFF	ON	OFF	OFF
38400	OFF	OFF	ON	OFF
19200	OFF	OFF	OFF	ON

SLAVE ID DESCRIPTION



For Slave ID Selection SW is used to Set The SLAVE ID .
For Slave ID DIP Switch LSB is "1" follow through "5" is MSB.
Slave ID Confirmed through below Device ID table .

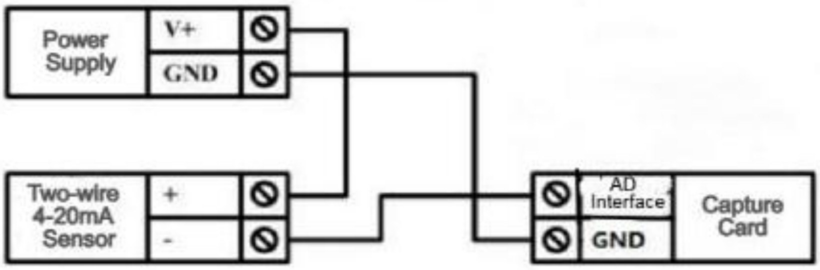
IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So"1 0 0 0" will be selected as Slave ID 1.

Note: DIP switches **SW1 to SW5** are provided for user configuration. These switches can be set as per the application requirements to control functions such as **slave address selection**, **mode settings**, or other device parameters.

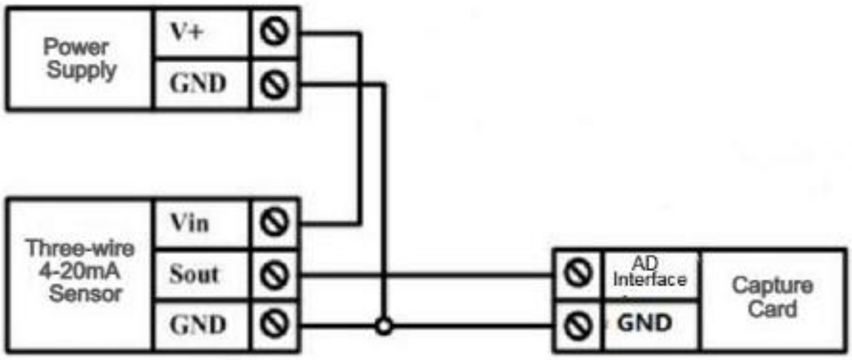
Note: The device supports configurable slave addresses within the range of **1 to 31**, which can be set according to application requirements.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

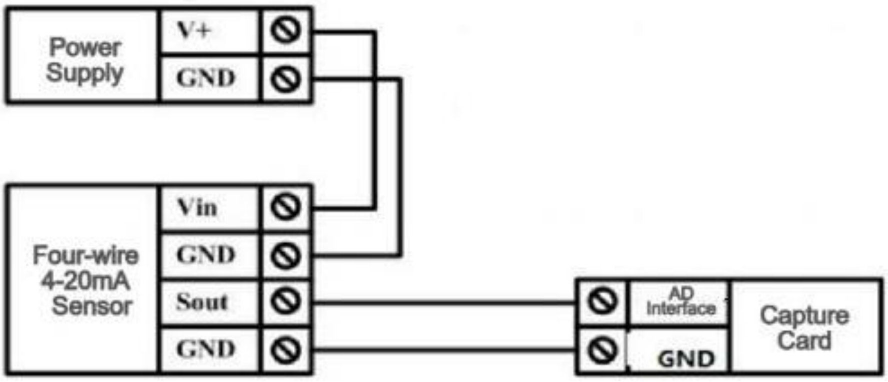
Two-wire Sensor Wiring Diagram



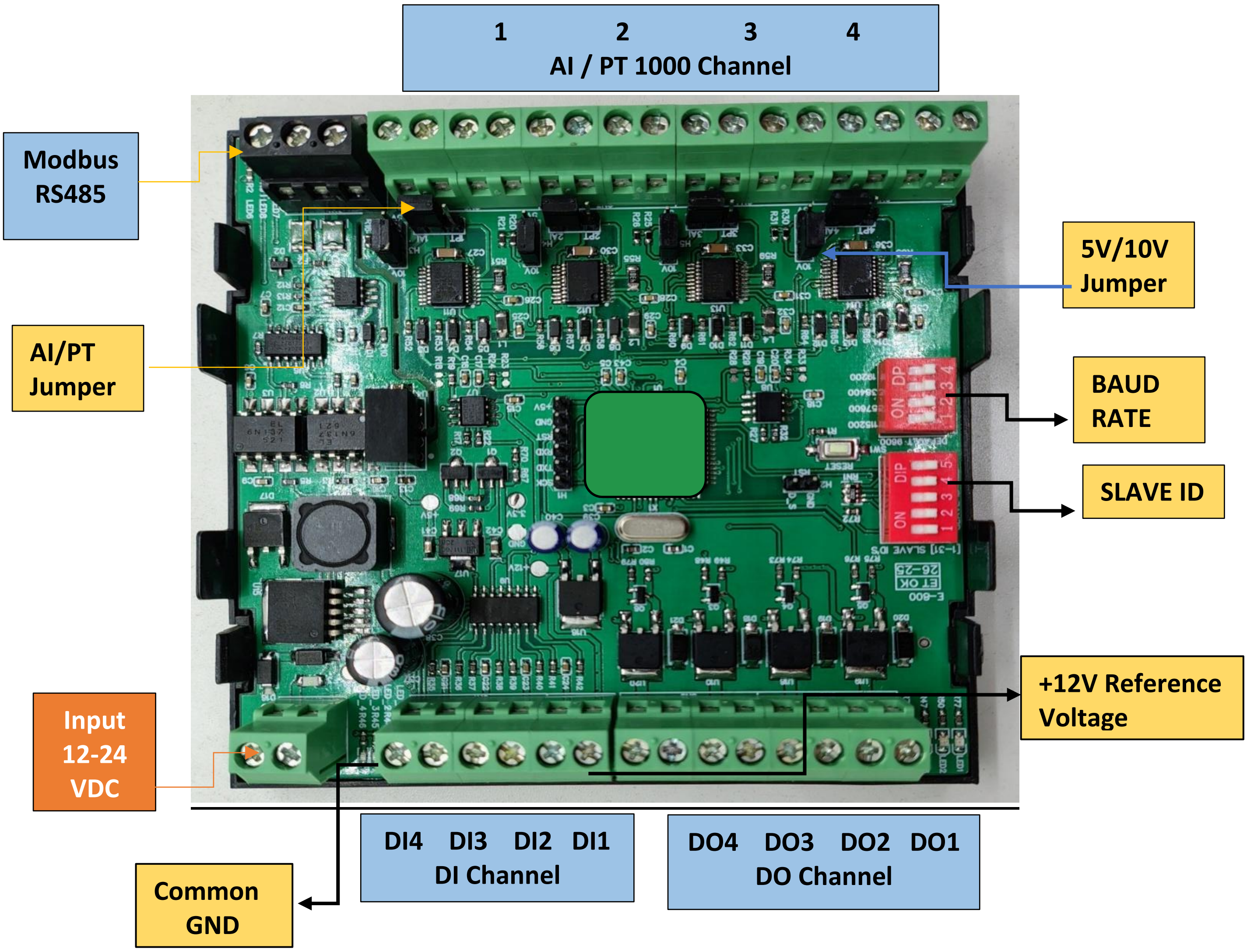
Three-wire Sensor Wiring Diagram



Four-wire 4-20mA Sensor



WIN – IO – 4DDAPM



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